one question: Is the overshoot you observe in a signal coming from the thing you are trying to measure, or from the filter you have used?

Finally, it is often desired that the upper half of the step response be symmetrical with the lower half, as illustrated in (e) and (f). This symmetry is needed to make the *rising edges* look the same as the *falling edges*. This symmetry is called **linear phase**, because the frequency response has a phase that is a straight line (discussed in Chapter 19). Make sure you understand these three parameters; they are the key to evaluating time domain filters.

Frequency Domain Parameters

Figure 14-3 shows the four basic frequency responses. The purpose of these filters is to allow some frequencies to pass unaltered, while completely blocking other frequencies. The **passband** refers to those frequencies that are passed, while the **stopband** contains those frequencies that are blocked. The **transition band** is between. A **fast roll-off** means that the transition band is very narrow. The division between the passband and transition band is called the **cutoff frequency.** In analog filter design, the cutoff frequency is usually defined to be where the amplitude is reduced to 0.707 (i.e., -3dB). Digital filters are less standardized, and it is common to see 99%, 90%, 70.7%, and 50% amplitude levels defined to be the cutoff frequency.

Figure 14-4 shows three parameters that measure how well a filter performs in the frequency domain. To separate closely spaced frequencies, the filter must have a **fast roll-off**, as illustrated in (a) and (b). For the passband frequencies to move through the filter unaltered, there must be no **passband ripple**, as shown in (c) and (d). Lastly, to adequately block the stopband frequencies, it is necessary to have good **stopband attenuation**, displayed in (e) and (f).

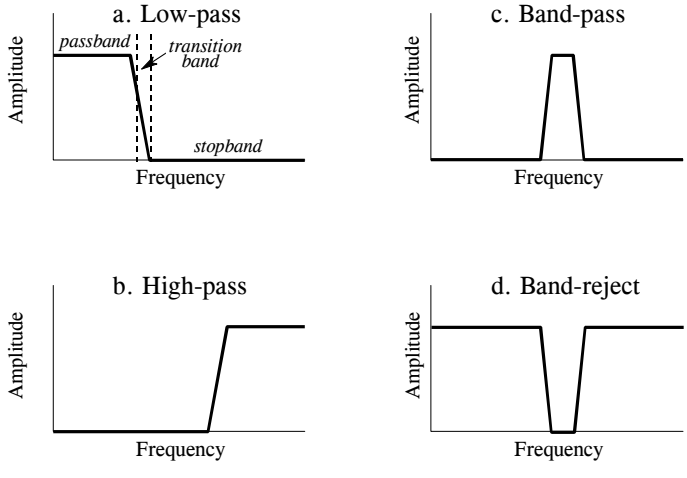

FIGURE 14-3
The four common frequency responses. Frequency domain filters are generally used to pass certain frequencies (the *passband*), while blocking others (the *stopband*). Four responses are the most common: low-pass, high-pass, band-pass, and band-reject.